SIGMIDI Foot Controller v0.1 — USB firmware installer

Use desktop **Chrome or Edge**, a USB-C data cable and the **Arduino Nano ESP32 / ESP32-S3, 16 MB flash**. This installer includes firmware and ESP Web Tools locally and works without internet. Web Serial requires HTTPS or localhost. iPad Safari can use the controller's Wi-Fi editor after installation; it cannot run this USB installer.

Open the installer

From the complete project, run `python3 scripts/serve_flasher.py` and open **http://localhost:8766/**. For an extracted installer ZIP, run this command inside its directory:

```
python3 -m http.server 8766 --bind 127.0.0.1
```

For static hosting, publish the whole directory over HTTPS, including `firmware/` and `vendor/`. Serve `.js` as JavaScript and `.bin` as binary. Relative paths support subdirectory hosting; missing assets must return an error, not the HTML index.

Install

1. Export a configuration backup from an operational controller.
2. Connect USB-C, jumper B1 to GND, tap RST, then remove the jumper. Select the new ROM serial port.
3. Choose **Connect & install**, select the controller and confirm.
4. Leave **Erase device** unchecked to keep compatible settings. Select erase for an intentional reset; it removes mappings, calibration and settings.
5. Wait for completion and restart. Tap RST if it remains in ROM mode. Repeat the B1/GND procedure for subsequent updates; the application exposes USB MIDI, not serial/DFU.

The page validates firmware hashes and compatibility before enabling installation. Opening the page does not write to a device.

Configure

The BLE name is **SIGMIDI Foot Controller v0.1**. On the controller, open **Wi-Fi Setup**, join the displayed network, open **http://192.168.4.1/** and use the displayed password. Configure the ladder count, calibrate inputs and assign MIDI actions.

This firmware uses passive pedal inputs with two resistors per controller input: 5.1 kΩ from 3V3 to RING and 10 kΩ from D2/GPIO5 or D3/GPIO6 to TIP. TIP connects directly to ADC A1/GPIO2 or A3/GPIO4; SLEEVE connects to GND. Resistor mode drives the bias GPIO HIGH; potentiometer and disabled modes leave it at high impedance. Recapture pedal heel/toe after rewiring. The external pedals are unpowered 0–10 kΩ variable resistors or potentiometers.

Choose **Left · Screen on left** for 180° display rotation and A–E / 1–5 list order from left to right. **Right · Screen on right** uses 0° and reverses each row. Left is the default. MIDI actions, names, calibration and LED assignments belong to physical taps. Save the selected variant.

Flash layout and integrity

The boot-region image contains the bootloader at `0x0` and partition table at `0x8000`, ending below `0x9000`. The application starts at `0x20000`. NVS at `0x9000–0x18fff` and PHY data at `0x19000–0x19fff` are outside the write images. Unchecked erase preserves these regions; firmware startup recovery can reset unusable NVS.

The installer checks manifest/image SHA-256 hashes, lengths, ESP32-S3 headers and safe flash ranges, then passes those verified bytes to ESP Web Tools. `SHA256.json` lists distributed-file hashes. They detect corruption and do not authenticate a publisher. The downloadable application alone needs a compatible

bootloader/partition layout.

ESP Web Tools **10.4.0** is distributed locally under Apache-2.0. See `vendor/esp-web-tools/LICENSE`, `vendor/esp-web-tools/provenance.json` and `vendor/THIRD_PARTY_NOTICES.txt`. Physical USB writing and restart require testing with a controller.

Native USB MIDI and BLE MIDI send the same PC/CC values when both links are ready. Add 100 kOhm from Nano VBUS to A4/GPIO11 and 150 kOhm from A4/GPIO11 to GND for USB detection while VIN-powered. Nano VIN accepts 6-21 V: use the positive 9 V barrel sleeve and common GND, not the Pololu 5 V output. The converter supplies the optional LEDs. Do not connect external 5 V to USB VBUS. Export any factory data before first installation: this installer replaces Arduino firmware/partitions and does not provide double-reset DFU. Compatible SIGMIDI NVS preservation applies after the first installation.